Technology Stack

Date	25 June 2026
Team ID	xxxxxx
Project Name	Smart Lender – Loan Approval Prediction System
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Technology Stack Details:

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** (e.g., React, Angular, HTML/CSS)	HTML5, CSS3	HTML5 and CSS3 are used to develop a responsive, simple, and user-friendly web interface for entering applicant details and displaying loan prediction results.

2	**Backend / Server-Side** (e.g., Node.js, Python/Django, Java)	Python (Flask)	Flask is a lightweight Python web framework used to build the web application. It loads the trained Machine Learning model and processes user input to generate loan approval predictions.
3	**Database / Data Storage** (e.g., MySQL, MongoDB, PostgreSQL)	Pickle(.pkl Files)	The trained Machine Learning model (model.pkl) and feature scaler (scaler.pkl) are stored using Python Pickle and loaded by the Flask application to perform loan predictions. No database is used in this project.
4	**Cloud / Hosting / Deployment** (e.g., AWS, Azure, Heroku, Docker)	Render	Render is used to deploy the Flask application online. It supports GitHub integration, automatic deployment, and provides public access to the application.

5	**Version Control & CI/CD** (e.g., GitHub, GitLab, Jenkins)	GitHub	GitHub is used for version control, source code management, collaboration, and maintaining the complete project repository.
6	**Third-Party APIs/ Other Tools** (e.g., Stripe, Twilio, Google Maps)	Google Colab, Kaggle Loan Prediction Dataset	Google Colab is used for developing, training, and evaluating the Machine Learning model. The Kaggle Loan Prediction Dataset is used for data preprocessing, visualization, model training, and testing.